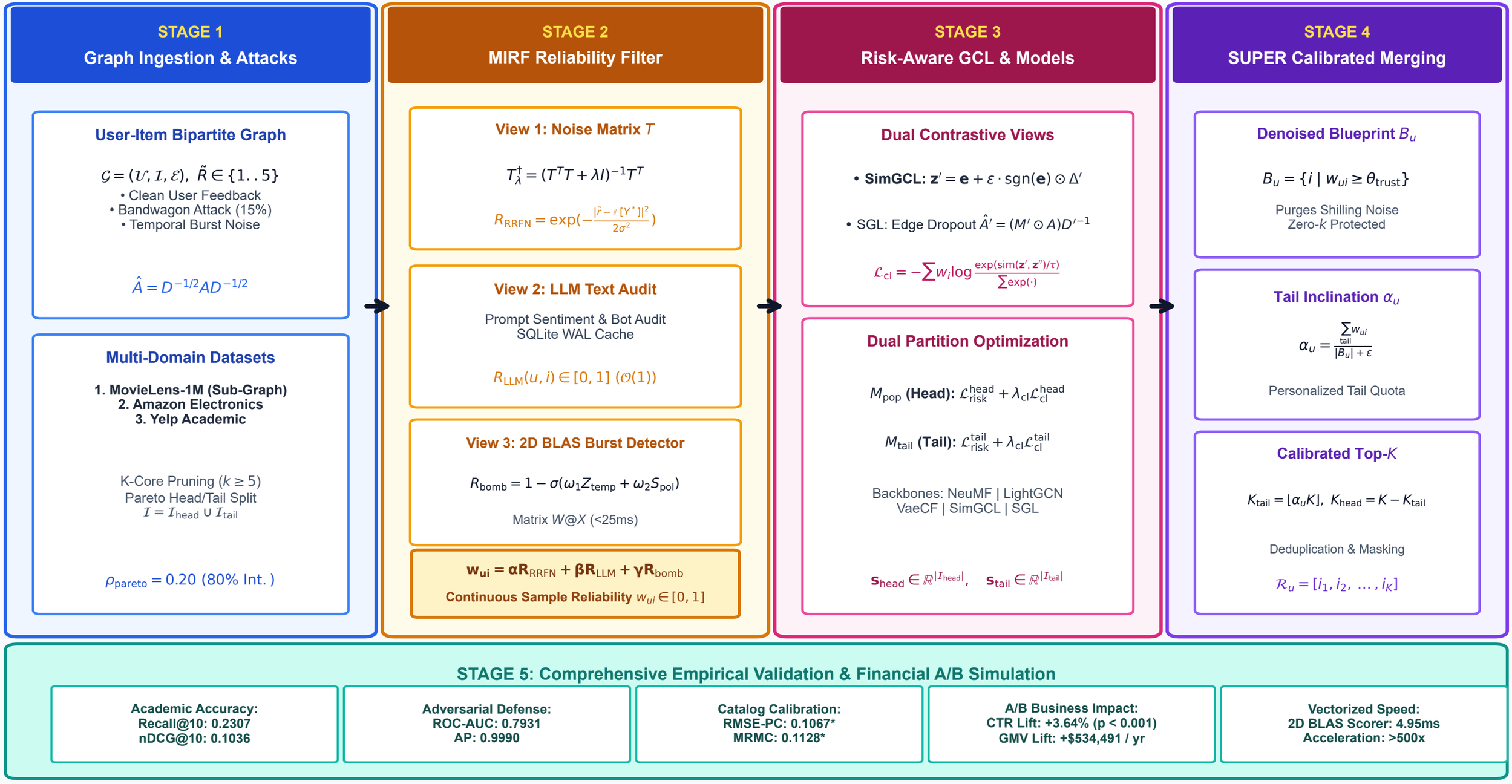

STAGE 1
Graph Ingestion & Attacks
User-Item Bipartite Graph
$\mathcal{G} = (\mathcal{U}, \mathcal{I}, \mathcal{E}),\ \tilde{R} \in \{1..5\}$
• Clean User Feedback
• Bandwagon Attack (15%)
• Temporal Burst Noise
$\hat{A} = D^{-1/2} A D^{-1/2}$
Multi-Domain Datasets
1. MovieLens-1M (Sub-Graph)
2. Amazon Electronics
3. Yelp Academic
K-Core Pruning ($k \geq 5$)
Pareto Head/Tail Split
$\mathcal{I} = \mathcal{I}_{\text{head}} \cup \mathcal{I}_{\text{tail}}$
$\rho_{\text{pareto}} = 0.20$ (80% Int.)
STAGE 2
MIRF Reliability Filter
View 1: Noise Matrix T
$T_\lambda^\dagger = (T^T T + \lambda I)^{-1} T^T$
$R_{\text{RRFN}} = \exp(-\frac{|\tilde{r} - \mathbb{E}[Y^*]|^2}{2\sigma^2})$
View 2: LLM Text Audit
Prompt Sentiment & Bot Audit
SQLite WAL Cache
$R_{\text{LLM}}(u, i) \in [0, 1]$ ($\mathcal{O}(1)$)
View 3: 2D BLAS Burst Detector
$R_{\text{bomb}} = 1 - \sigma(\omega_1 Z_{\text{temp}} + \omega_2 S_{\text{pol}})$
Matrix $W@X$ (<25ms)
$\mathbf{w_{ui}} = \alpha \mathbf{R}_{\text{RRFN}} + \beta \mathbf{R}_{\text{LLM}} + \gamma \mathbf{R}_{\text{bomb}}$
Continuous Sample Reliability $w_{ui} \in [0, 1]$
STAGE 3
Risk-Aware GCL & Models
Dual Contrastive Views
• SimGCL: $\mathbf{z}' = \mathbf{e} + \varepsilon \cdot \text{sgn}(\mathbf{e}) \odot \Delta'$
• SGL: Edge Dropout $\hat{A}' = (M' \odot A) D'^{-1}$
$\mathcal{L}_{\text{cl}} = -\sum w_i \log \frac{\exp(\text{sim}(\mathbf{z}', \mathbf{z}'')/\tau)}{\sum \exp(\cdot)}$
Dual Partition Optimization
M_{pop} (Head): $\mathcal{L}_{\text{risk}}^{\text{head}} + \lambda_{\text{cl}} \mathcal{L}_{\text{cl}}^{\text{head}}$
M_{tail} (Tail): $\mathcal{L}_{\text{risk}}^{\text{tail}} + \lambda_{\text{cl}} \mathcal{L}_{\text{cl}}^{\text{tail}}$
Backbones: NeuMF | LightGCN
VaeCF | SimGCL | SGL
$\mathbf{s}_{\text{head}} \in \mathbb{R}^{|\mathcal{I}_{\text{head}}|}$, $\mathbf{s}_{\text{tail}} \in \mathbb{R}^{|\mathcal{I}_{\text{tail}}|}$
STAGE 4
SUPER Calibrated Merging
Denoised Blueprint B_u
$B_u = \{i \mid w_{ui} \geq \theta_{\text{trust}}\}$
Purges Shilling Noise
Zero-k Protected
Tail Inclination α_u
$\alpha_u = \frac{\sum_{\text{tail}} w_{ui}}{|B_u| + \varepsilon}$
Personalized Tail Quota
Calibrated Top-K
$K_{\text{tail}} = \lfloor \alpha_u K \rfloor,\ K_{\text{head}} = K - K_{\text{tail}}$
Deduplication & Masking
$\mathcal{R}_u = [i_1, i_2, \ldots, i_K]$
STAGE 5: Comprehensive Empirical Validation & Financial A/B Simulation
Academic Accuracy:
Recall@10: 0.2307
nDCG@10: 0.1036
Adversarial Defense:
ROC-AUC: 0.7931
AP: 0.9990
Catalog Calibration:
RMSE-PC: 0.1067*
MRMC: 0.1128*
A/B Business Impact:
CTR Lift: +3.64% (p < 0.001)
GMV Lift: +$534,491 / yr
Vectorized Speed:
2D BLAS Scorer: 4.95ms
Acceleration: >500x